CatBoost 1.6 1.5 0.26 0.36 0.27 0.29 0.36
DeepONet 0.84 1.4 0.27 0.4 0.22 0.27 0.36
Domain-informed Concat-MLP 1 1.8 0.23 0.27 0.22 0.23 0.32
Domain-informed FiLM-MLP 1 1.5 0.19 0.24 0.16 0.18 0.27
Extra Trees 1.5 1.4 0.32 0.42 0.29 0.34 0.34
F-FNO 2.2 4.1 0.37 0.52 0.4 0.45 0.4
GAMS-GNN-style 1.3 1.8 0.3 0.46 0.27 0.35 0.38
Latent Neural Operator 1.3 2.2 0.26 0.4 0.22 0.25 0.35
Mean baseline 6.5 6.9 2.9 2.6 0.71 0.78 0.36
PINN continuity 1.2 1.3 0.27 0.43 0.24 0.32 0.36
PINN data-only 1.2 1.7 0.25 0.33 0.23 0.25 0.33
Random Forest 1.6 1.5 0.24 0.31 0.22 0.25 0.33
Ridge 5 8.9 0.4 0.62 0.75 0.77 0.36
Transolver-style 1.3 1.9 0.3 0.45 0.27 0.35 0.38
U-FNO3D 1.8 3.5 0.37 0.52 0.34 0.42 0.39
XGBoost 1.4 1.6 0.22 0.31 0.21 0.24 0.32
Density
Temperature
TED
TKE
u
v
w
Mean median case-normalized MAE over three seeds
1
2
3
4
5
6
7
8